North South University

Department of Electrical & Computer Engineering

Fall 2025

Course Title: Analog Electronics I Lab

Faculty Name: Ms. Syeda Sarita Hassan [SSH1]

Section: 7

Experiment Number: 04

Experiment Name:

Clipper and Clamper Circuits

Experiment Date: 04-11-2025

Date of Submission: 11-11-2025

Group Number: 01

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Experiment name : Clipper and Clamper Circuits

Objective :

(1) To understand the working principles of clipper and clamper circuits using semiconductor diodes.

(2) To observe how clipper circuits limit or remove specific portions of an input AC signal above or below a certain voltage.

(3) To learn how clamper circuits shift the DC level of an AC waveform without changing its overall shape.

(4) To varify the theoritical behavior of clipper and clamper circuits through experimental observation using an oscilloscope.

Theory:

Clipper: A clipper is a circuit that clips off the magnitude of voltage of a signal without affecting any other part of the signal. It removes parts of the signals waveform that exceed a given voltage level. A clipper either clips the specific part or the entire positive or negative half cycle of the signal.

A diode connected in series with the load can clipped off any half cycle of input depending on the orientation of the diode.

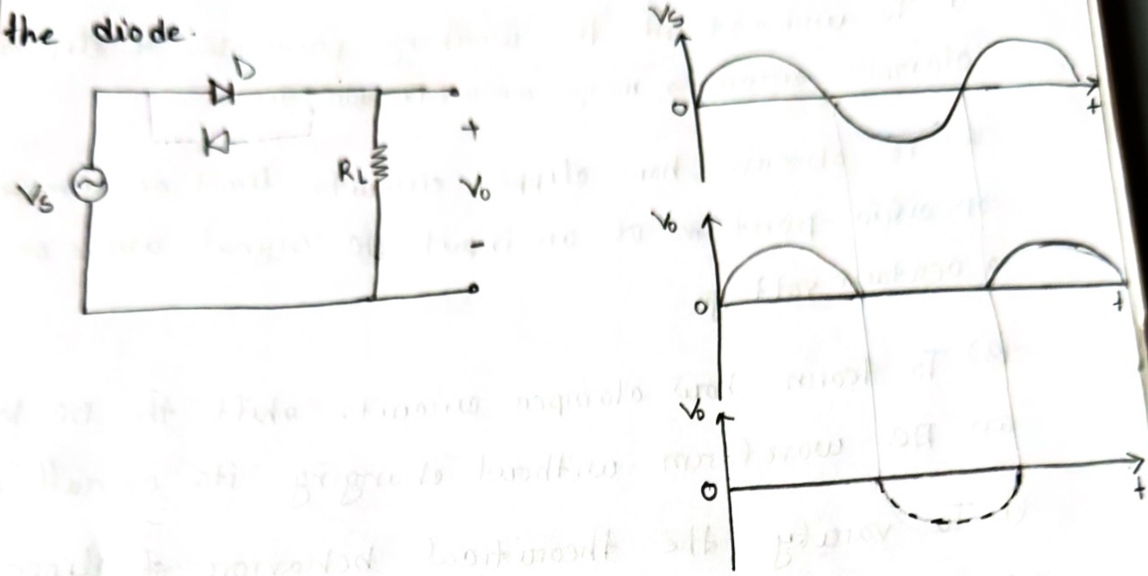

Figure: Simple Diode Clipper Circuit

If a voltage source (battery) of V volts in reverse bias condition with the diode is added to the circuit, then for V_s above $(V+0.7)$ volts the diode becomes forward bias and turns on. The load only receives the portion of the signal above this voltage threshold.

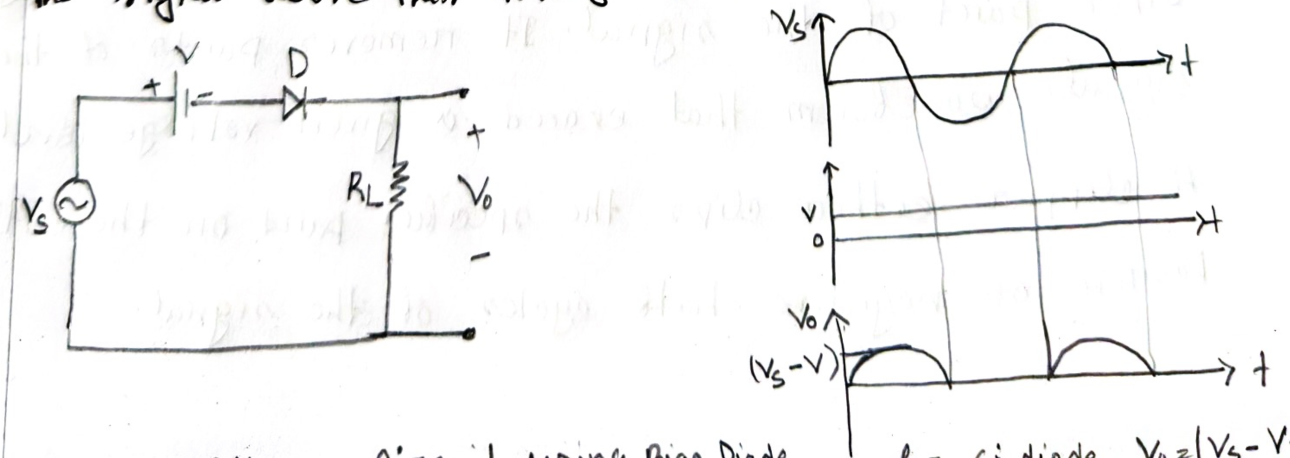

Figure: Clipper Circuit using Bias Diode.

for Si diode, $V_0 = (V_s - V - 0.7V)$

If a diode is connected in parallel to the load, it can remove the part of the input signal above 0.7 volts during one half cycle, based on how the diode is oriented. Placing two diodes in parallel facing opposite directions, both half cycle can be clipped to a maximum of 0.7 volts.

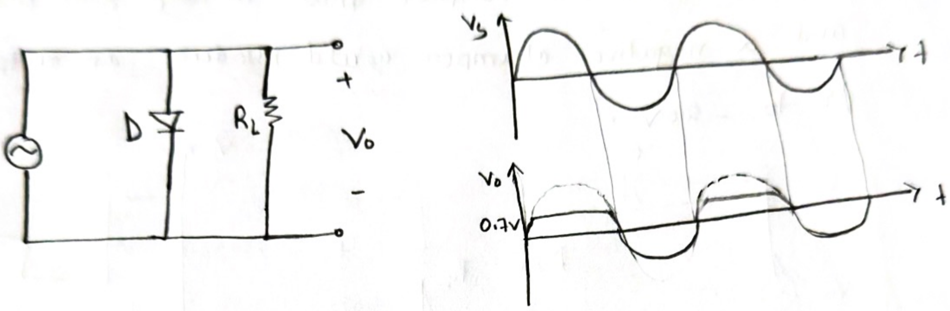

Figure: Parallel Clipper Circuit.

A biased diode clips the output voltage to a set level based on the battery. Either half or both hal of the signal can be clipped.

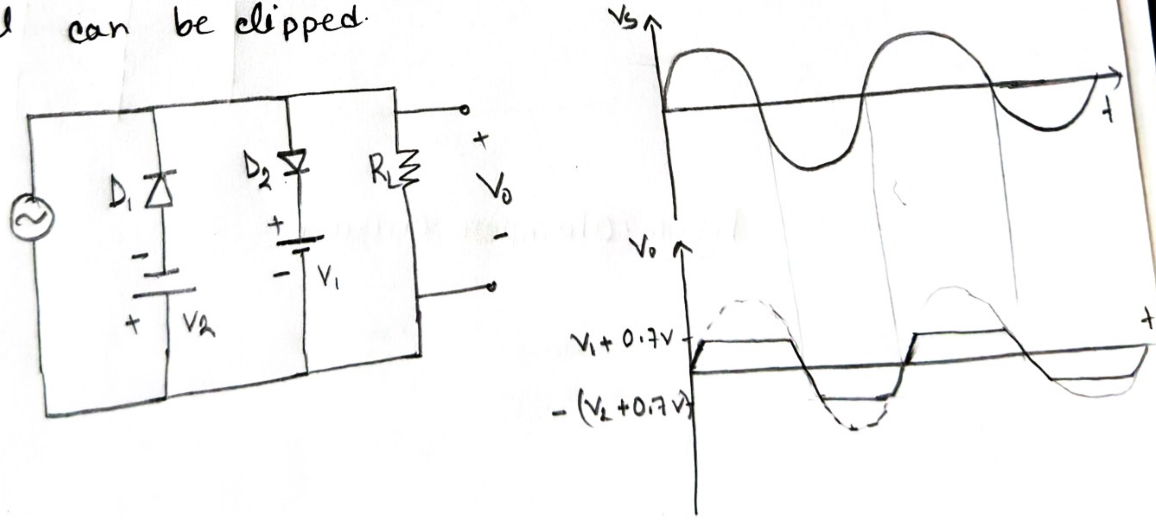

Figure: Biased Parallel Clipper Circuit

Clamper : A clamper circuit is an electronic circuit that adds a DC level to an AC signal, shifting the entire waveform without altering its shape. For example, if the input signal varies from -10 v. to 10v , a positive DC clamper will produce an output that ideally goes from 0 to 20 v and a negative clamper could produce an output between 0v to -20V .

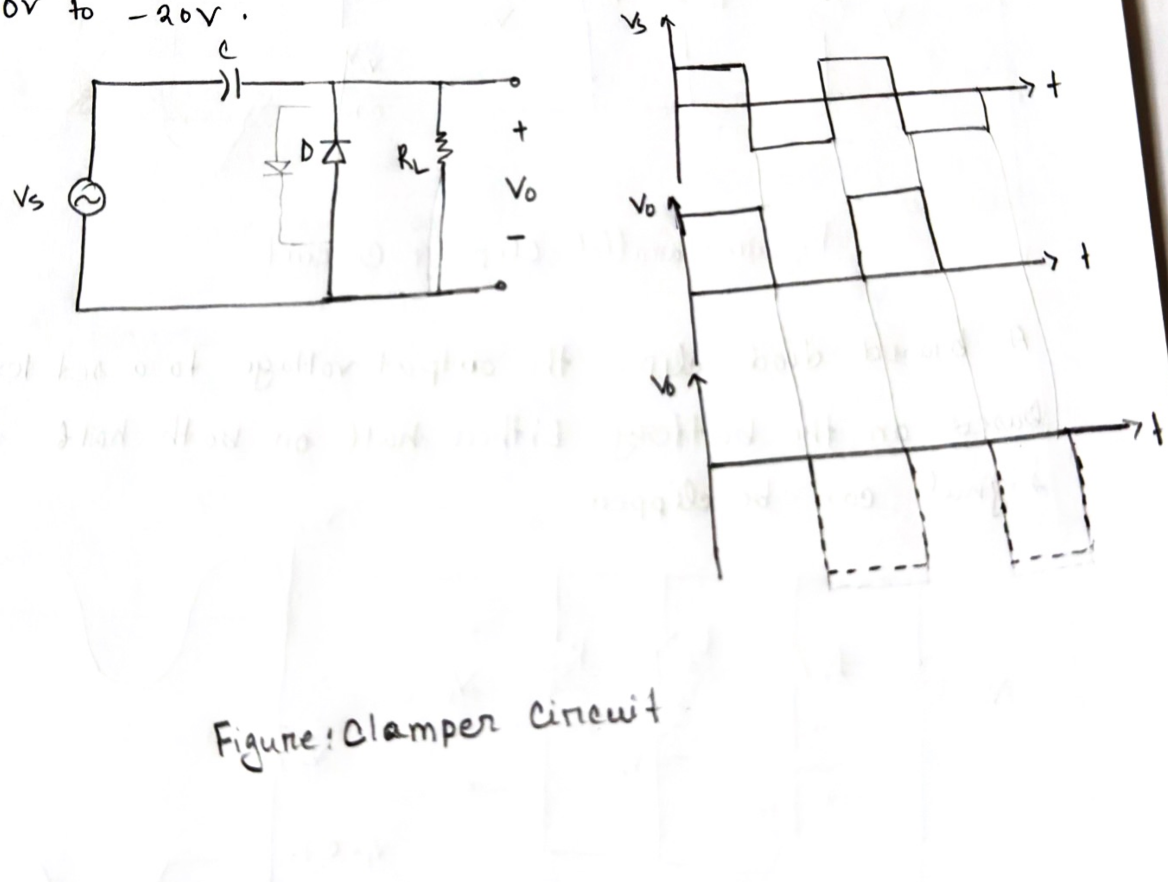

Figure: Clamper Circuit

List of Equipment :

Serial no.	Component Details	Specification	Quantity
1.	P-n Junction diode	1N4007	1 Piece
2.	Resistor	100 kΩ	1 Piece
3.	Capacitor	0.1 μF	1 Piece
4.	Signal generator		1 unit
5.	Trainer Board		1 unit
6.	DC Power Supply		1 unit
7.	Oscilloscope		1 unit
8.	Digital Multimeter		1 unit
9.	Chords and wire		as required

Circuit Diagram:

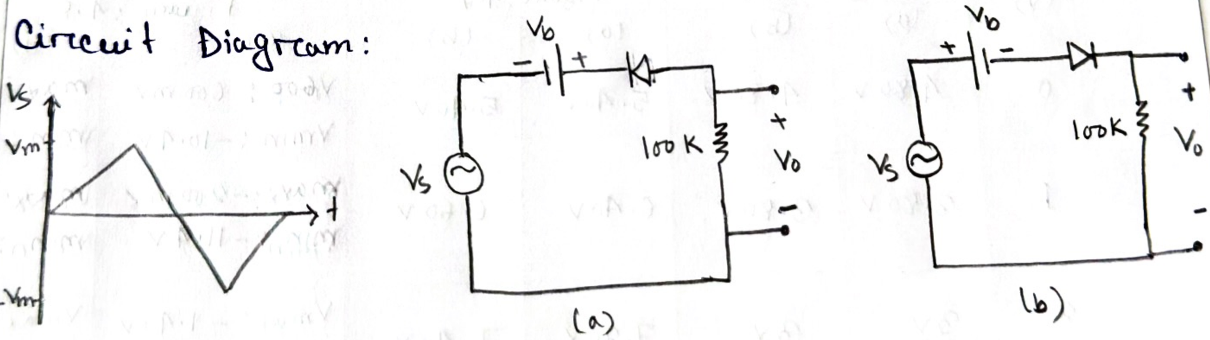

Figure 4.6: Series Cipper Circuit

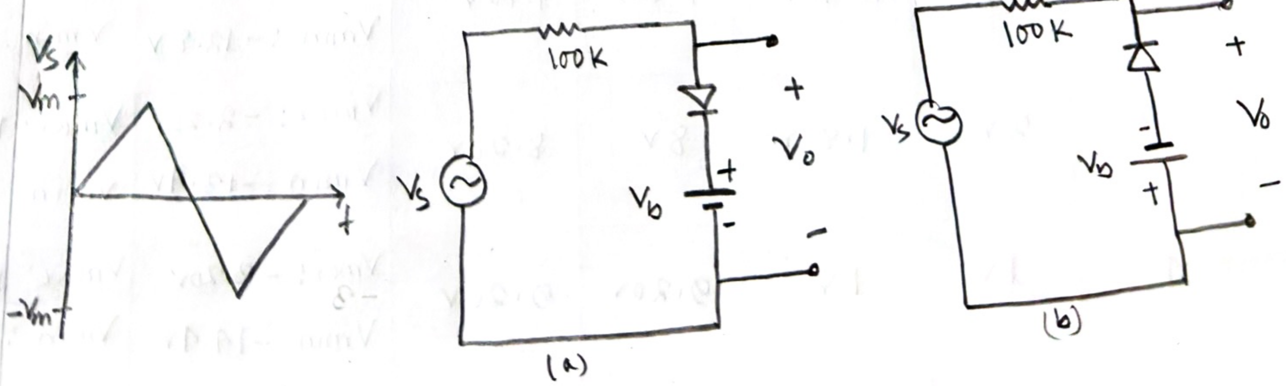

Figure 4.7: Parallel Clipper Circuit

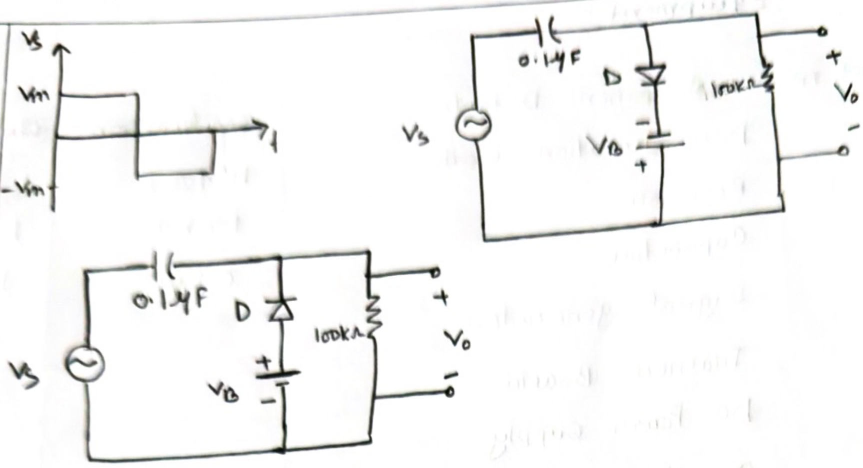

Figure 4.8: Clamper circuit

Data Collection:

Vb (V)	Vo (P-P) Figure: 4.6 (a)	Figure: 4.6 (b)	Figure: 4.7 (a)	Figure: 4.7 (b)	Figure: 4.8 (a)	Figure: 4.8 (b)
0	4.80 V	4.80 V	5.40 V	5.40 V	Vmax: 600 mV Vmin: -10.4 V	Vmax: 9.80 V Vmin: -800 mV
1	3.80 V	3.80 V	6.40 V	6.60 V	Vmax: -200 mV Vmin: -11.4 V	Vmax: 10.8 V Vmin: 200 mV
2	3 V	3 V	7.20 V	7.40 V	Vmax: -1.40 V Vmin: -12.2 V	Vmax: 12 V Vmin: 1 V
2.5	2.40 V	2.40 V	7.80 V	7.80 V	Vmax: -2 V Vmin: -12.8 V	Vmax: 12.4 V Vmin: 1.60 V
3	2 V	1.80 V	8 V	8.20 V	Vmax: -2.20 V Vmin: -13.4 V	Vmax: 12.8 V Vmin: 2 V
4	1 V	1 V	9.20 V	9.20 V	Vmax: -3.20 V Vmin: -14.4 V	Vmax: 13.8 V Vmin: 3 V
5	0.2 V	0.2 V	9.40 V	9.40 V	Vmax: -4.40 V Vmin: -15.4 V	Vmax: 15 V Vmin: 3.80 V

Graph:

Q:01 (Am)

Roll No. ....

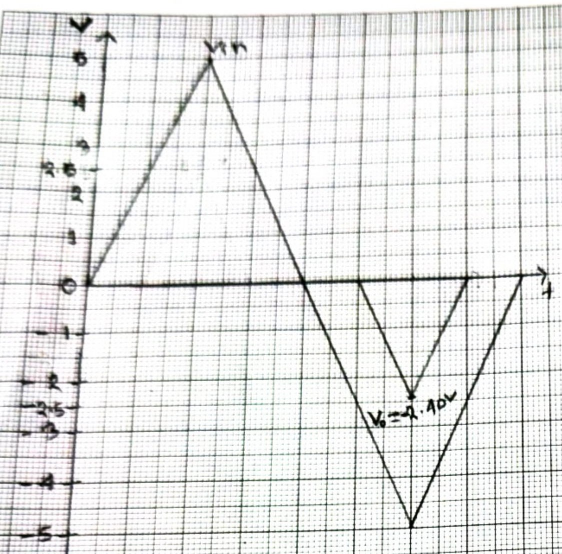

Fig: 4.6 (a)

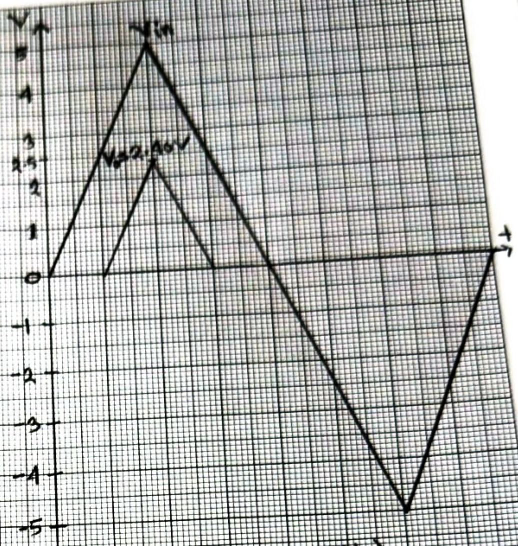

Fig: 4.6 (b)

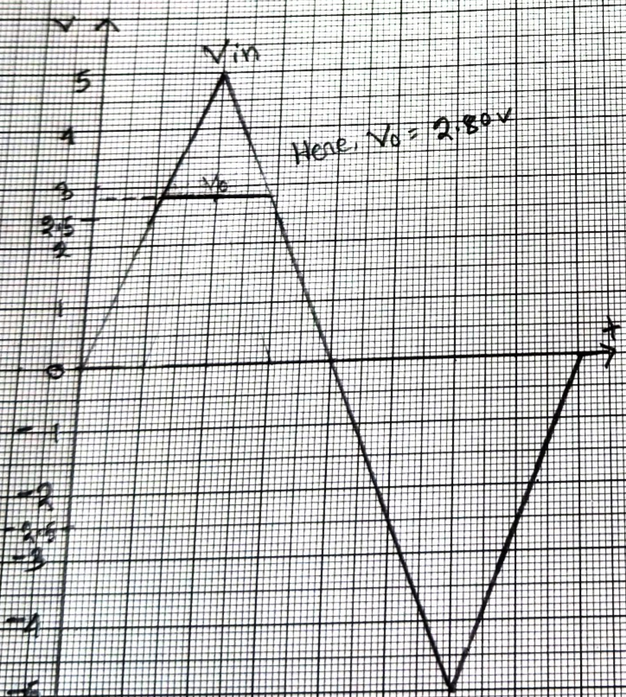

Fig: 4.7 (a)

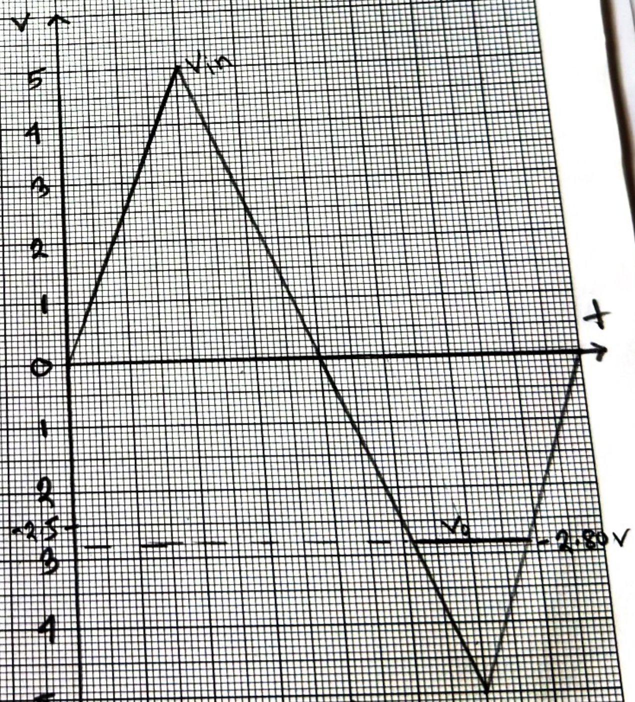

Fig: 4.7 (b)

20 cm x 25 cm

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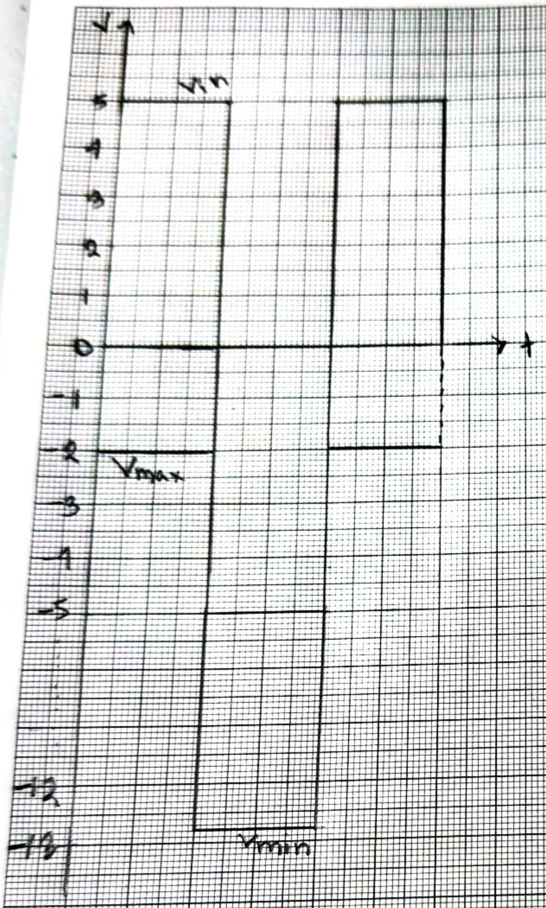


Fig 4.8(a)

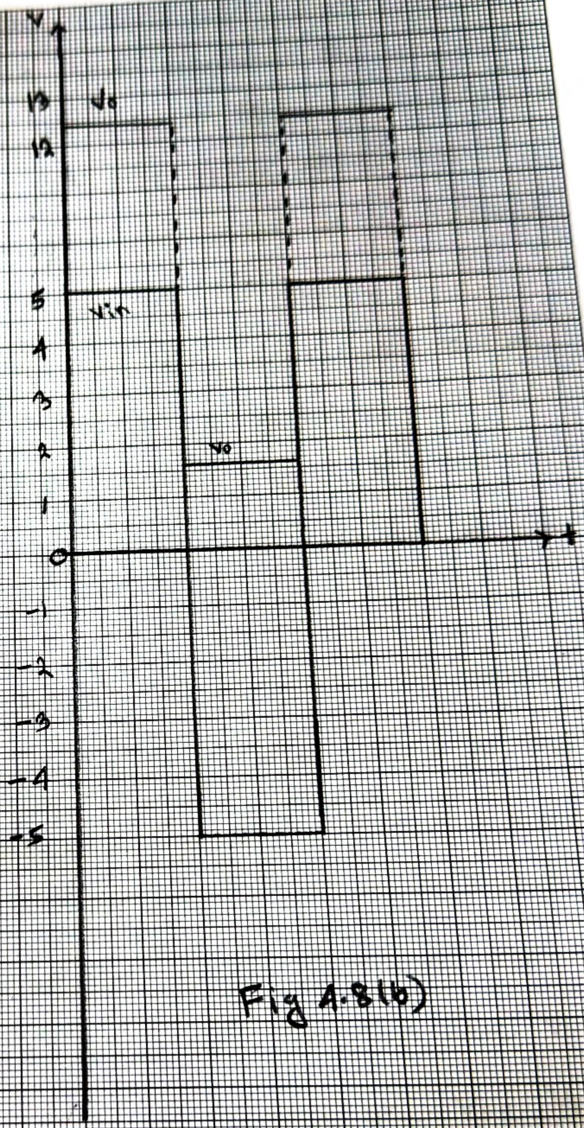


Fig 4.8(b)

KE 20 cm x 25 cm

Question and answer:

Q-2: In procedure 4, we have decreased the value of V_b from 2.5V to 0V. In the series clipper circuit (Fig 4.6(a) and 4.6(b)), we observed that the value of V_o (P-P) is increasing. That means decreasing the value of V_b increases the value of V_o.

But in parallel clipper circuit (Fig. 4.7(a) and 4.7(b)), we observed that, the value of V_o decreases as V_b decreases. When V_b is 2.5 V, V_o is 7.80 V. It starts decreasing from 7.80V to 5.40V as we decreased the V_b from 2.5V to 0V.

In both cases, diode turns on $V_{in} > V_b$. In a series clipper, the diode and bias voltage are placed in the path of the output, so output signal is only present when the diode is in forward biased. In contrast, in a parallel clipper, the diode is in parallel with the load, the output is present when diode is turned off. In this case, $V_o = V_s$ or $V_o = V_{in}$. When the diode is turned on, $V_o = V_b + 0.7V$.

But in series clipper circuit, $V_o = 0$ when the diode is off. So, decreasing V_b, lowers the knee voltage, allowing the diode to pass more of the signal in series clipper. But in parallel clipper circuit, the diode clips more when V_b decreases.

In case of clamper circuit, we got halfware signal in one direction for $V_b = 2.5V$. As clamper circuit shifts the entire waveform in one direction. So, $V_o(P-P)$ remain almost same despite decreasing V_b.

Q-03: In procedure 5, we have increased the value of V_b from 2.5V to 5V. In case of series clipper circuit (Figure 4.6(a), 4.6(b)), we observed that $V_o(P-P)$ decreases as we increase V_b.

But in parallel clipper circuit (Fig 4.7(a), 4.7(b)), V_o increases as V_b increases.

In series clipper circuit, increasing V_b cause the increase in knee voltage. So the diode need more voltage to turned on, resulting the decrease in output voltage.

In parallel circuit, the diode gives more output signal when V_b increases. As when the diode is on and V_b increases, $V_o = V_b + 0.7V$. So V_o increases as well.

In clamper circuit, V_o remain same when V_b increases. It only shift the wavefrom in one direction.

Discussion:

In this experiment, we have learned about clipper and clamper circuits and how they works. Here, we also observed how the output signal varries under the influence of bias voltage. We learned about series and parallel clippers and how they are oriented. We also observed, how clamper circuit shifts the DC level of an AC signal up or down.

The experimental data and graph were almost same as we predicted from theoritical knowledge. We observed that in series clipper circuit Vo decreases as the bias voltage increases. and in parallel clipper circuit Vo increases when bias voltage increases. Clampers shifts the DC levels without altering the signal's shape.

In this experiment, we faced difficulties dealing with the oscilloscope. However the readings and waveforms were stable throughout the experiment.

In conclussion, the experiment successfully demonstated the theoritical concepts of clippers and clampers circuit.

NORTH SOUTH UNIVERSITY
DEPARTMENT OF ELECTRICAL & COMPUTER ENGINEERING

Signature of instructor: [signature] 9-11-2025

Data Collection:

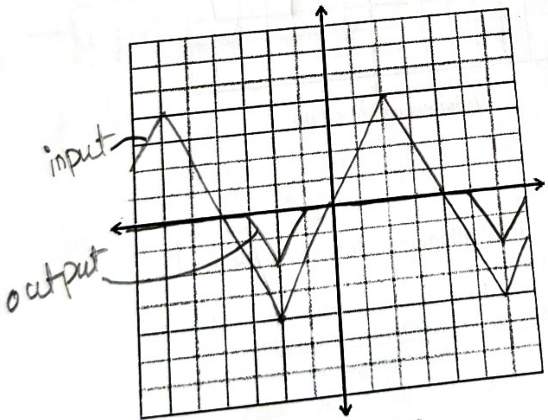
Experiment: 4,
Performed by Group# 1

Theoretical value: R = 100 kΩ

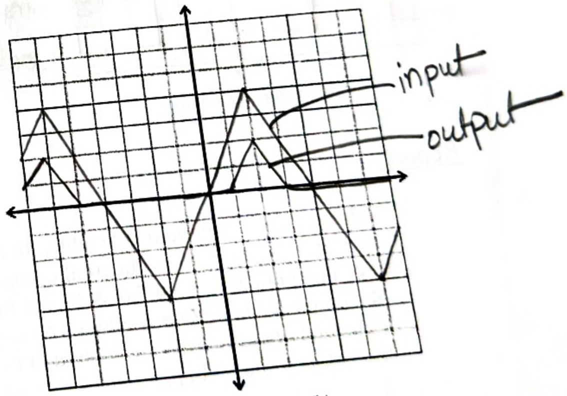
Measured value: R = 98.6 kΩ

Vs = 10.6 V(p-p).

V_b (V)	Vo (p-p)					
	Fig 4.6		Fig 4.7		Fig 4.8	
	(a)	(b)	(a)	(b)	(a)	max (b) min
0	4.80V	4.80V	5.40V	5.40V	600mV, -10.4V	9.80V, -800mV
1	3.80V	3.80V	6.40V	6.60V	-200mV, -11.4V	10.8V, 200mV
2	3V	3V	7.20V	7.40V	-1.40V, -12.2V	12V, 1V
2.5	2.40V	2.40V	7.80V	7.80V	V_{max} = -2V, V_{min} = -12.8V	V_{max} = 12.4V, V_{min} = 1.60V
3	2V	1.80V	8V	8.20V	-2.20V, -13.4V	12.8V, 2V
4	1V	1V	9.20V	9.20V	-3.20V, -14.4V	13.8V, 3V
5	200mV	200mV	9.40V	9.40V	-4.40V, -15.4V	15V, 3.80V



Input-output Fig 4.6 (a)



Input-output Fig 4.6 (b)

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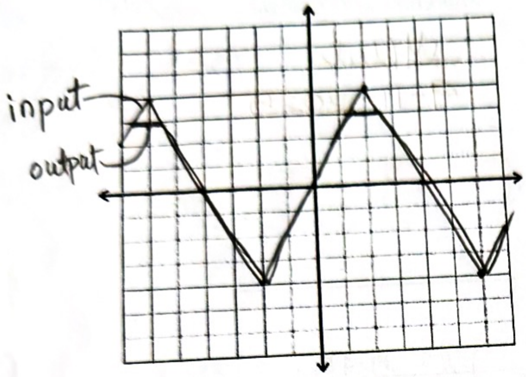
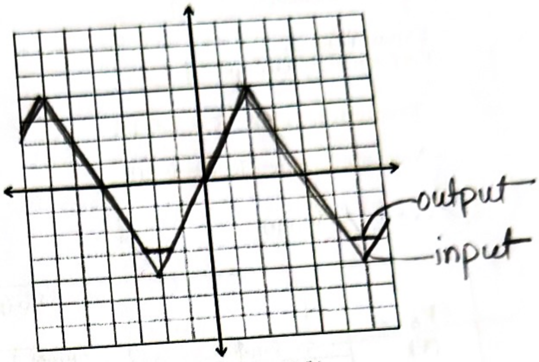
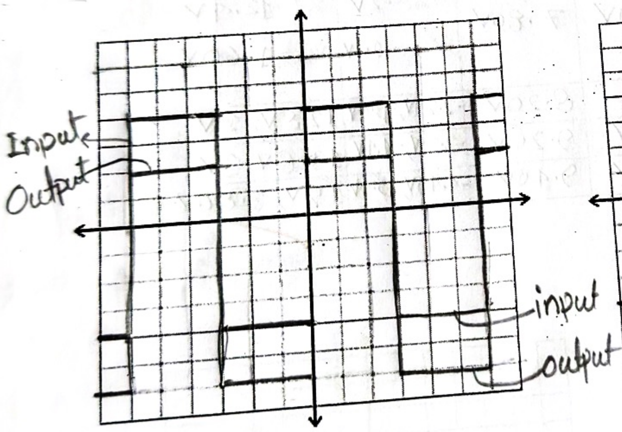


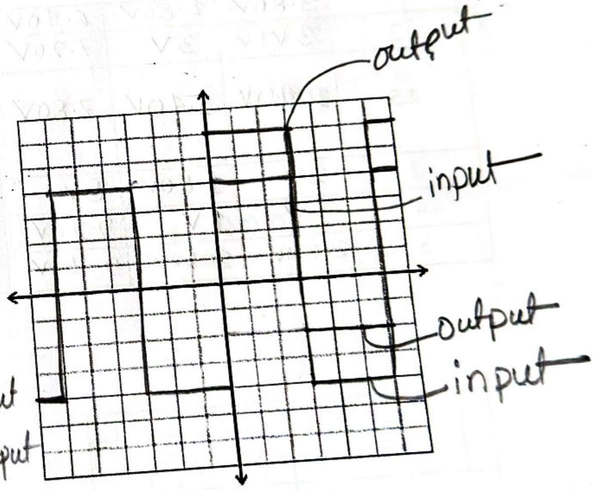
Input-output Fig 4.7 (a)



Input-output Fig 4.7 (b)



Input-output Fig 4.8 (a)



Input-output Fig 4.8 (b)

<u>Report:</u>

1. Using values from your data table, for all the circuit diagrams plot the input-output waveforms observed on the oscilloscope for V_b=2.5V.
2. For Fig 4.6(a &b), Fig4.7 (a & b) and Fig 4.8 (a & b) what change did you observe in the output voltage, In procedure-4? *<u>Explain the reason behind such a change.</u>*
3. For Fig 4.6(a &b), Fig4.7 (a & b) and Fig 4.8 (a & b) what change did you observe in the output voltage, In procedure-5? *<u>Explain the reason behind such a change.</u>*